# Asynchronous Physical Exchanges

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## Introduction

Today, technologies based on blockchain make it possible to carry out digital exchanges without a trusted third party, relying on cryptography and decentralized systems such as Bitcoin.

However, as soon as an exchange involves a physical good, trust becomes necessary again because the digital world and the material world cannot be directly merged.

This document proposes a reflection around a conceptual process aimed at reducing certain risks related to asynchronous physical exchanges, that is to say exchanges of material goods between distant parties, not simultaneously present and not knowing each other, without claiming to resolve all existing issues.

## 1. Definition of the problem

### 1.1 General context

The exchange of physical goods at a distance is today largely facilitated by digital networks. However, when two parties do not know each other and are not present at the same time, these exchanges still rely largely on mechanisms of human trust, intermediation, or centralized institutions.

### 1.2 Asynchronism as the main source of risk

The fundamental problem appears when the exchange is not simultaneous. The buyer and the seller interact neither in the same place nor at the same moment. This absence of simultaneity creates an asymmetry of risk that cannot be eliminated by a simple agreement between the parties. Asynchronism transforms a simple exchange into a problem of coordination and trust.

### 1.3 Risk perceived by the seller

From the seller's point of view, the main risk is to relinquish a physical good without a certain guarantee of receiving the corresponding payment. Once the object is shipped, the seller loses all control over it. In the absence of irreversible payment or a guarantee mechanism, the seller is exposed to a definitive loss.

### 1.4 Risk perceived by the buyer

From the buyer's point of view, the risks are multiple. The first is to never receive the object after having made the payment. The second is to receive an object that does not

correspond to the expected description. Finally, even in the case of effective reception, the buyer may fear that the object is defective, unusable, or altered, without having an immediate and objective means to prove it.

### 1.5 Limits of current solutions

The solutions currently used rely either on centralized institutions, or on platforms playing the role of a trusted intermediary. These approaches may reduce certain risks, but introduce other constraints: dependence on a third party, loss of anonymity, arbitrary procedures, delays, or asymmetry of power.

### 1.6 Boundary between the digital world and the physical world

The core of the problem lies in the difficulty of making the digital world and the physical world interact reliably. A payment can be verified cryptographically, but a physical object cannot be entirely digitized nor guaranteed by purely mathematical proof. There is no perfect cryptographic proof of the state of a physical object.

### 1.7 Framework and intention of the proposed process

The process described in the following sections does not claim to resolve all these issues. Its objective is more modest: to reduce certain asymmetries of risk during asynchronous physical exchanges, relying on observable physical data and on cryptographic mechanisms, while preserving the anonymity of the parties.

This process must be understood as a line of reflection and not as a definitive solution.

## 2. Conceptual framework

### 2.1 Actors, role of the process

Three main actors are considered:

A (Buyer): wishes to acquire a physical good without being present at the same moment and in the same place as the seller. A seeks to reduce the risk of paying without receiving.

V (Seller): wishes to sell a physical good without being present at the same moment and in the same place as the buyer. V seeks to reduce the risk of shipping without being paid.

L (Logistics): ensures the physical transport between two points. L is not considered as a trusted arbitral third party regarding the content of the exchange, but as an operator subject to operational constraints and potentially economic responsibility in the case of non-delivery or anomaly detected upon arrival.

The process relies on a deposit/withdrawal infrastructure (lockers) allowing observable physical characteristics to be recorded during deposit and compared upon reception.

## 3. Overview of the process

### 3.1 General principle

This section presents an overall view of the sequence of an asynchronous physical exchange as envisaged by the proposed process. The objective is to describe the general sequence of events, without yet going into the specific technical details.

The process relies on two distinct but complementary layers:

· A settlement layer (Layer A), ensured by a truly decentralized and censorship-resistant blockchain, such as Bitcoin (BTC), used exclusively for irreversible payments.

· A state and coordination layer (Layer B), ensured by a distributed cryptographic infrastructure linked to physical lockers, making it possible to manage deposits, validations, delays and withdrawals, while preserving the anonymity of the parties.

### 3.2 Initiation of the exchange

An exchange begins when a buyer (A) and a seller (V) agree on the sale of a physical good, without being simultaneously present and without knowing each other. This agreement is carried out via a software interface (for example an application or a dedicated service), closely linked to the infrastructure of Layer B, in order to guarantee consistency between the digital exchange and the physical lockers.

At this stage:

-No payment is made.

-No object is transferred.

An exchange identifier is created, serving as a common cryptographic reference between the buyer, the seller and the object concerned, without revealing their civil identities.

### 3.3 Deposit of the object by the seller

The seller goes to a physical locker assigned to them by the system.

During the deposit:

1. A deposit session is opened using the exchange identifier.

2. The seller places the object in the locker.

3. The locker automatically records observable physical characteristics.

4. These data are cryptographically associated with the session and broadcast via Layer B.

The locker then locks in a non-final state (“red” state):

· The object is immobilized.

· No payment is triggered.

· The seller retains the possibility of recovering their good in the event of refusal or expiration.

A cryptographic signature by the seller is required only in the event of recovery of the object, in order to prove that the depositor is indeed the legitimate entity associated with the exchange session and to prevent any attempt at impersonation or fraudulent recovery.

**3.4 Transmission and analysis by the buyer**

The physical data recorded during the deposit are transmitted to the buyer via Layer B.

The buyer may then:

· Examine these data.

· Compare them with what they expect to receive.

· Decide to accept or refuse the exchange.

A maximum time limit is deliberately introduced in order to avoid any hostage-taking of the process by one of the parties.

**3.5 Validation by the buyer and definitive locking**

When opening the exchange session in the state layer (layer B), the seller registers the public Bitcoin address to which the payment is to be sent.

After examining and accepting the physical characteristics observed at the time of deposit, the buyer proceeds with payment on the settlement layer (layer A) by sending the agreed amount to this address. Once the transaction has been broadcast on the Bitcoin network, the buyer sends the corresponding transaction ID to layer B.

This identifier allows layer B to publicly verify, by consulting the Bitcoin blockchain ledger, the existence of the payment and its correspondence with the address registered by the seller and the amount expected for the exchange.

Once this proof of payment has been verified in layer B, the deposit locker changes to a final state, preventing the seller from retrieving the item.

Throughout this phase, the locker remains locked in a non-final state for a predefined period, ensuring that the seller cannot retrieve the item while the buyer is verifying and paying.

So:

· The creation and broadcast of a payment transaction on Layer A (Bitcoin blockchain).

· The transition of the locker to a definitive state (“green” state).

· The impossibility for the seller to recover the object because the buyer’s payment transaction is triggered.

The buyer’s signature therefore irreversibly links:

· The payment.

· The physical locking.

· The finalization of the exchange.

**3.6 Refusal or expiration**

If the buyer explicitly refuses the exchange, or if the validation period expires without action:

· No payment is triggered.

· The locker remains in a non-final state.

· The seller may return to recover their object.

The recovery is carried out by means of a cryptographic proof issued from Layer B, linked to the initial exchange session and verified locally by the locker infrastructure. This proof makes it possible to authorize the opening of the locker without identity disclosure, while guaranteeing that only the legitimate depositor may recover the object.

**3.7 Logistics intervention**

Once the locker is definitively locked following the buyer’s validation, the object becomes eligible for logistical handling. From that moment, the exchange between the buyer and the seller is considered finalized from a financial standpoint, the payment having been triggered on the settlement layer.

At the time of handling, the object is removed from the deposit locker by the logistics operator, who at that moment creates a cryptographic commitment, and is then transported to the planned reception location. It is important to note that, within the framework of this process, the object is supposed to be transported without any possible modification of its state, this state having been immobilized during the initial deposit by the seller and validated by the buyer on the basis of observable physical characteristics.

Upon arrival, the object is deposited in a reception locker. This locker then performs a new acquisition of observable physical characteristics and compares them with the data recorded during the initial deposit and already accepted by the buyer.

Any significant divergence between the characteristics measured upon reception and those recorded during the initial deposit indicates that the state of the object has been modified during logistical handling, even though it was supposed to remain unchanged since its initial immobilization. In this case, the deposit may be automatically refused by the system.

This verification requires no additional validation by the seller or the buyer, since they have already accepted the characteristics of the object at the moment of the initial deposit. In the event of conformity, the deposit is accepted, the locker moves to a valid state and the object becomes recoverable by the buyer.

In the event of non-delivery, deadlines being exceeded, or non-conformity detected upon reception, logistics may be held economically liable, according to automated settlement mechanisms relying on the settlement layer based on a decentralized public blockchain such as Bitcoin. This liability is intended to guarantee the integrity of the process without introducing an arbitral third party or calling into question the anonymity of the parties.

The access of logistics operators to the lockers is not based on general authority, but on temporary cryptographic capabilities issued by the coordination layer, strictly limited in time, space and purpose.

### 3.8 Recovery of the object by the buyer

Once the object has been delivered and validated by the reception locker, it becomes recoverable by the buyer. At this stage, the exchange is considered finalized from both a financial and logistical standpoint: the payment has already been triggered during the initial validation by the buyer, and the object has been transported without modification of its state.

Recovery by the buyer is carried out via a cryptographic proof linked to the exchange identifier and managed by the state and coordination layer (Layer B). This proof allows the buyer to open the corresponding reception locker, without revealing their civil identity or interacting with logistics or the seller.

During recovery, no new validation is required. The physical characteristics of the object having already been verified upon reception by the locker and compared with the data recorded during the initial deposit, the system considers the object to be compliant. The locker then moves to a single open state allowing the buyer to remove the object, and then closes definitively.

Once the object has been recovered, the exchange session is closed. No subsequent action is possible on this exchange via the infrastructure, which guarantees the finality of the process and prevents any reversibility or late dispute.